Project Seminar 2026





Design of a fuel cell monitoring board in KiCad and development of the operational board software

Lesson 8: Project work → PCB

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FIW: Fachbereich Elektrotechnik und Informatik

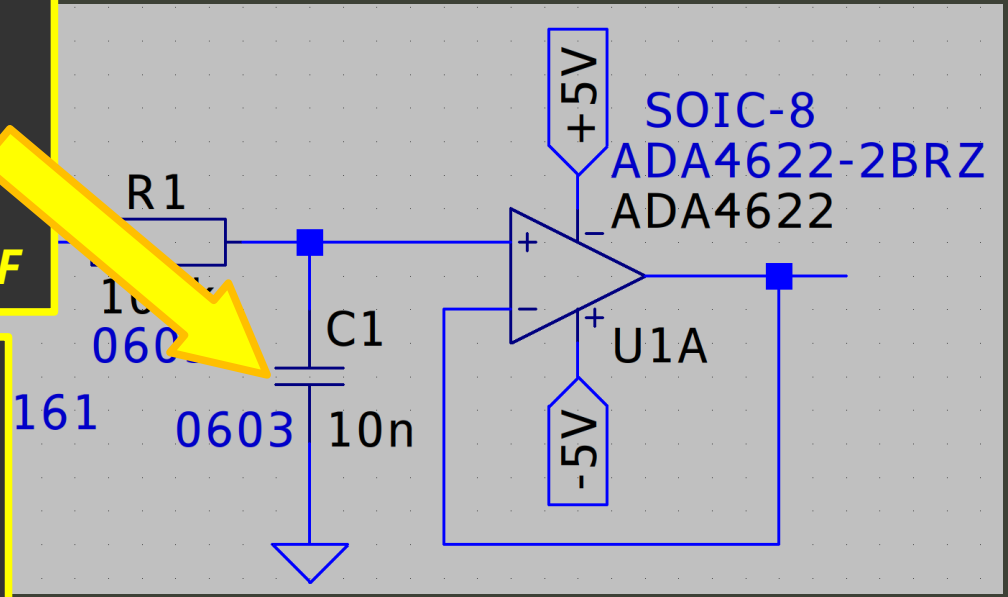
HOCHSCHULE WISMAR
UNIVERSITY OF APPLIED SCIENCES TECHNOLOGY, BUSINESS AND DESIGN

Project Seminar – Project Work

- ***Repetition on the topic "Basics PCB Design" (→ lecture 3):***

Annotation & value example C1 (100nF):

- ***C → Reference designator***
- ***1 → id number***
- ***100n → capacitance (value) 100nF***



Also possible:

- ***R15A, R15B***
  → resistor network
- ***U1A, U1B***
  → dual operational amplifier

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- ***Repetition on the topic "Reading Data Sheets":***

→ *2 Operational Amplifiers in one package!*

→ *2 supply pins (+Vs, -Vs)*
→ *4 input pins*
→ *2 output pins*

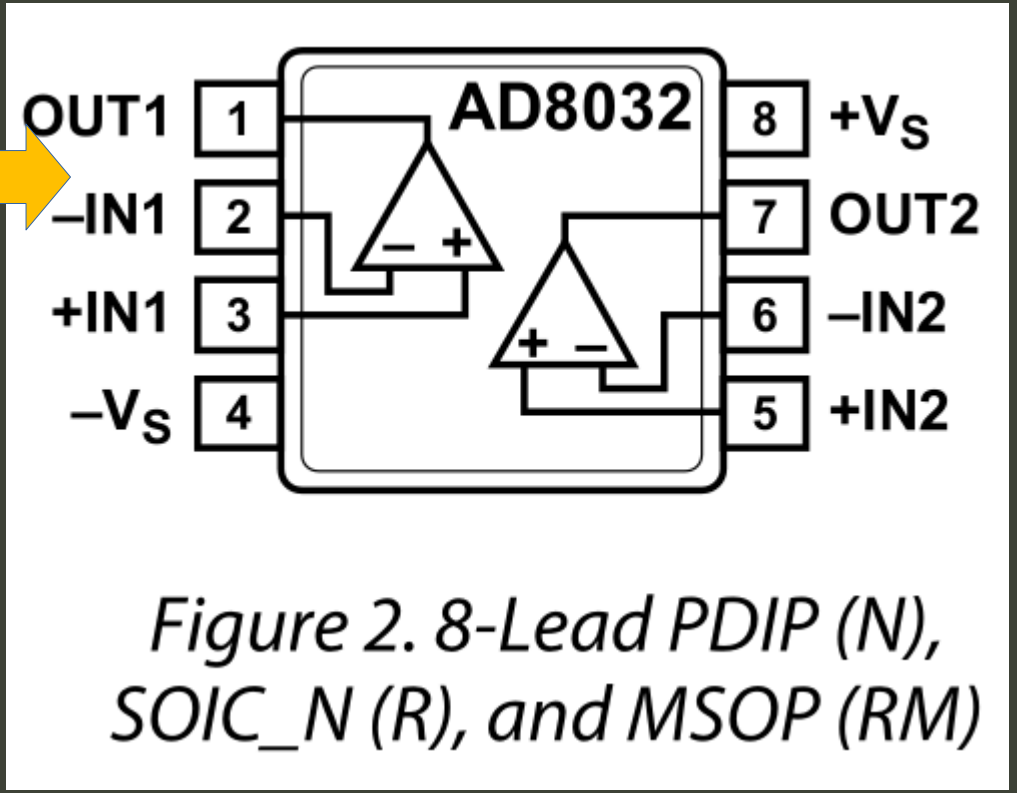


Figure 2. 8-Lead PDIP (N), SOIC_N (R), and MSOP (RM)

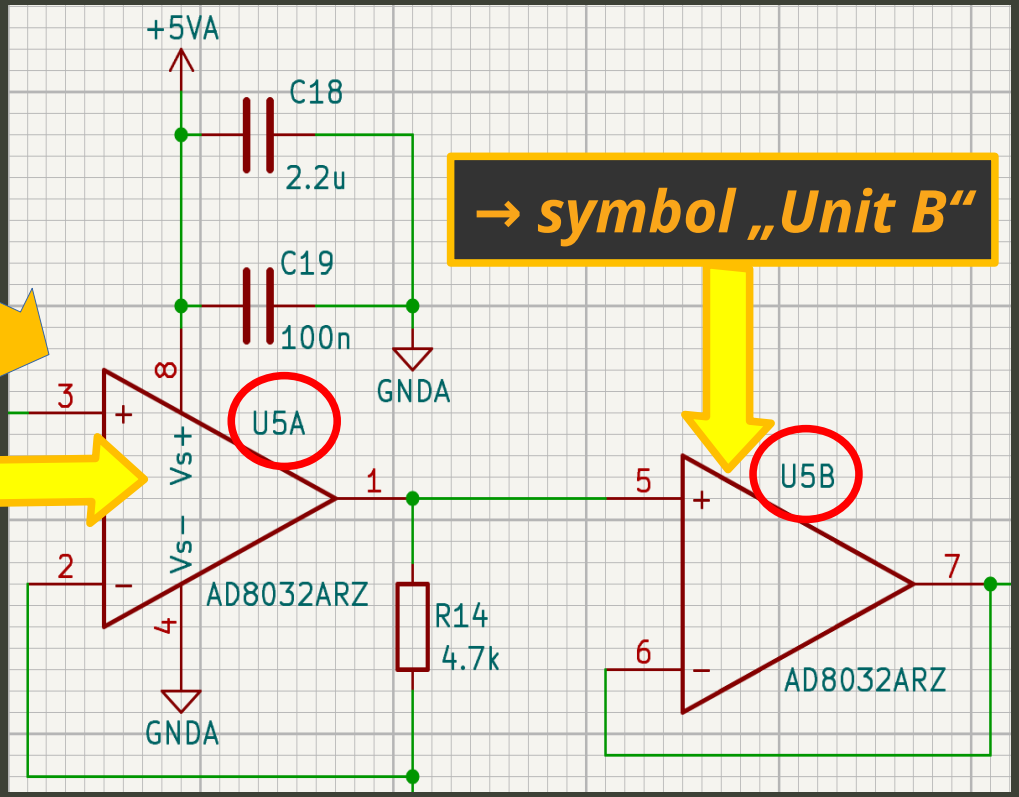
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- ***Repetition on the topic "creating symbols":***

→ *2 Operational Amplifiers in one symbol file!*

→ *symbol „Unit A"*

→ *symbol „Unit B"*



Project Seminar – Project Work





- ***Repetition PCB layer stackup → multi-layer design (F.Cu & B.Cu 70µm):***

Board Setup

- Board Stackup
  - Board Editor Layers
  - Physical Stackup
  - Board Finish
  - Solder Mask/Paste
  - Zone Hatch Offsets
- Text & Graphics
  - Defaults
  - Formatting
  - Text Variables
- Design Rules
  - Constraints
  - Pre-defined Sizes
  - Teardrops
  - Length-tuning Patterns
  - Tuning Profiles
  - Net Classes
  - Component Classes
  - Custom Rules
  - Violation Severity
- Board Data
  - Embedded Files

Copper layers: 6 | Impedance controlled | Add Dielectric Layer... | Remove Dielectric Layer...

Layer	Id	Type	Material	Thickness	Color	Epsilon R	Loss Tan
F.Silkscreen	Top Silk Screen		Not specified		Not specified		
F.Paste	Top Solder Paste						
F.Mask	Top Solder Mask		Not specified	0.01 mm	Not specified	3.3	0
F.Cu	Copper			0.07 mm			
Dielectric 1	PrePreg		FR4	0.1 mm			
In1.Cu	Copper			0.03			
Dielectric 2	Core		FR4	0.535 m			
In2.Cu	Copper			0.035 mm			
Dielectric 3	PrePreg		FR4	0.1 mm			
In3.Cu	Copper			0.035 mm			
Dielectric 4	Core		FR4	0.535 m			
In4.Cu	Copper			0.0			
Dielectric 5	PrePreg		FR4	0.1			
B.Cu	Copper			0.07 mm			
B.Mask	Bottom Solder Mask		Not specified	0.01 mm	Not specified	3.3	0
B.Paste	Bottom Solder Paste						
B.Silkscreen	Bottom Silk Screen		Not specified		Not specified		

Board thickness from stackup: 1.67 mm | Adjust Dielectric Thickness | Export to Clipboard

Import Settings from Another Board... | Abbrechen | OK

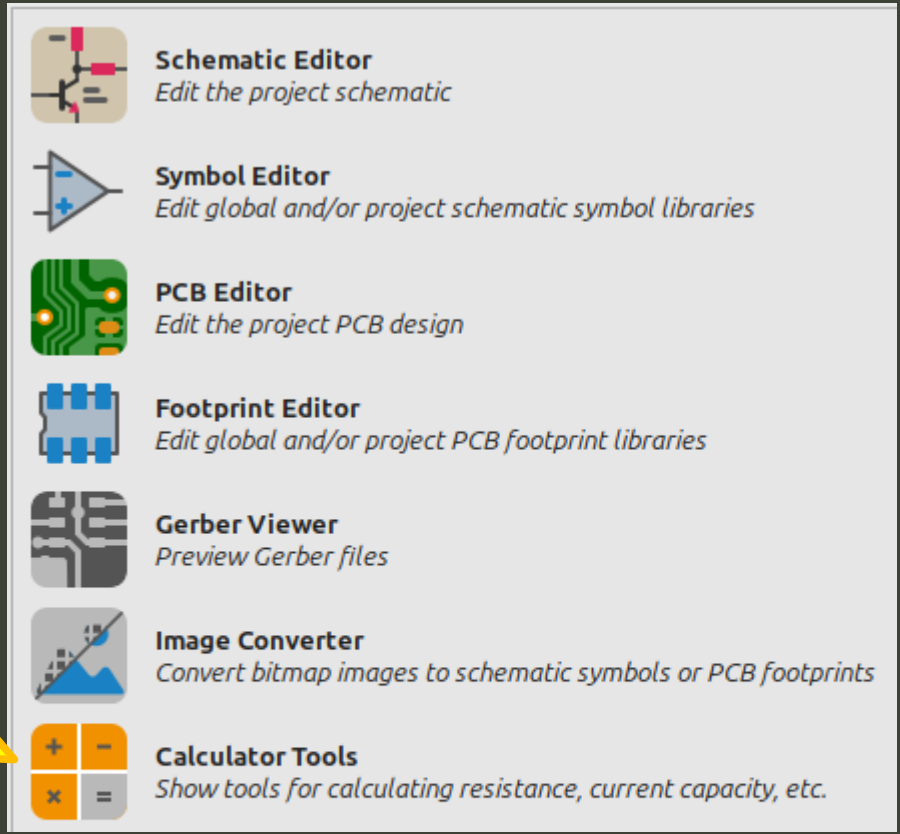
→ The layers F.Cu & B.Cu are configured for high current!

→ Thickness=70µm

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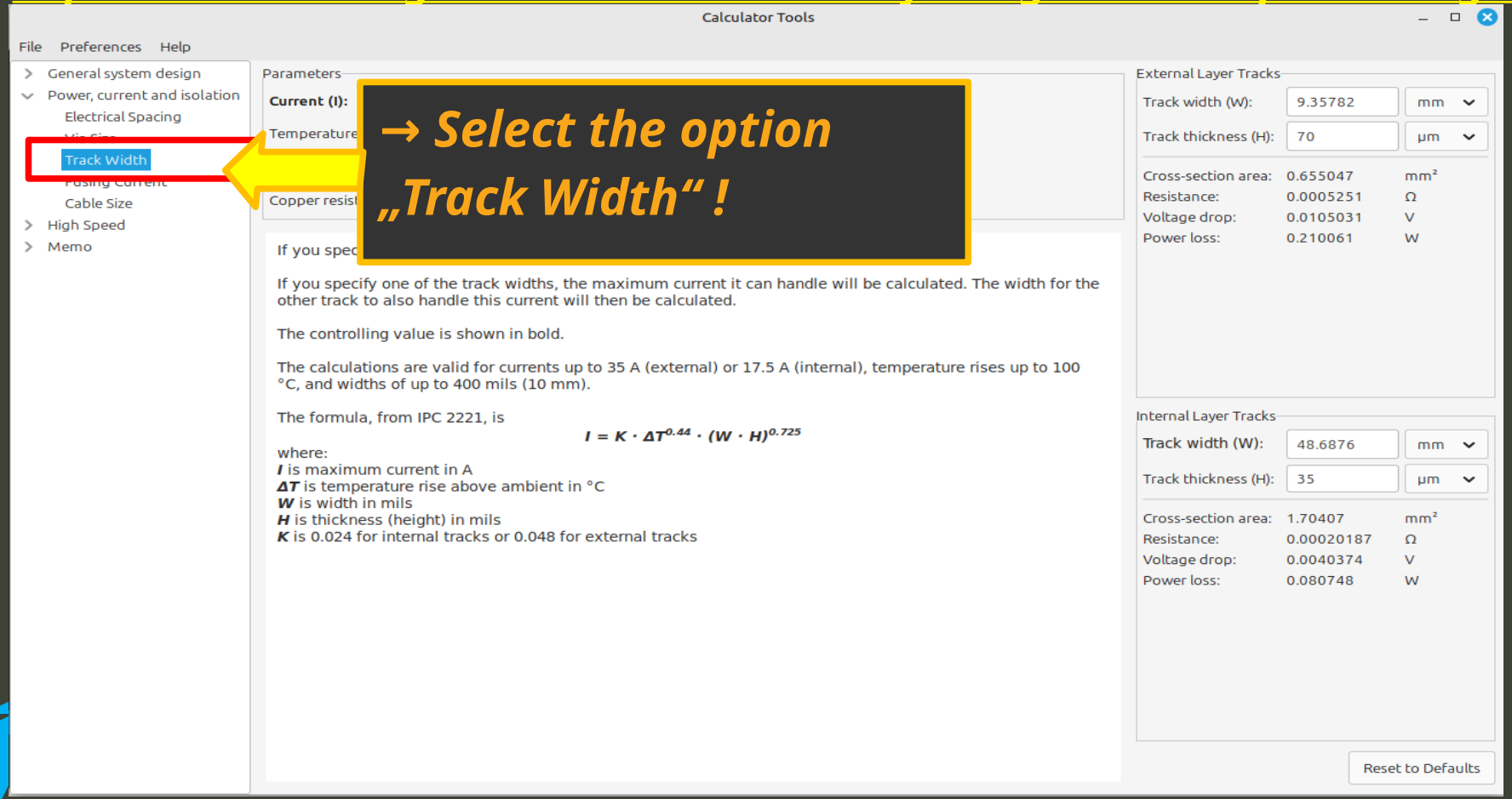
- ***Repetition: → Using KiCad calculator tool for high-current path design***

KiCad calculator tools in KiCad navigator window



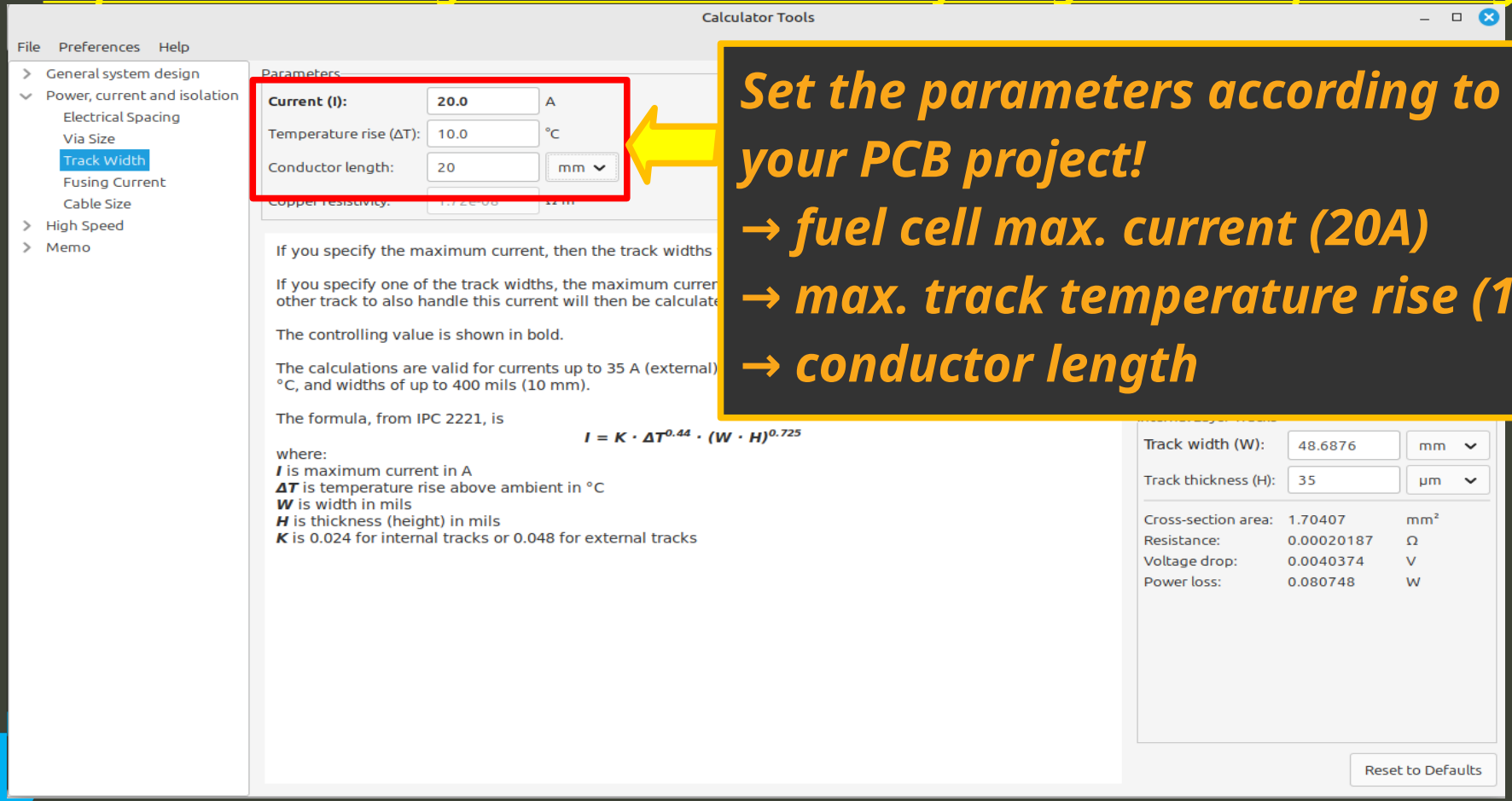
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- ***Repetition: → Using KiCad calculator tool for high-current path design***



Project Seminar – Project Work

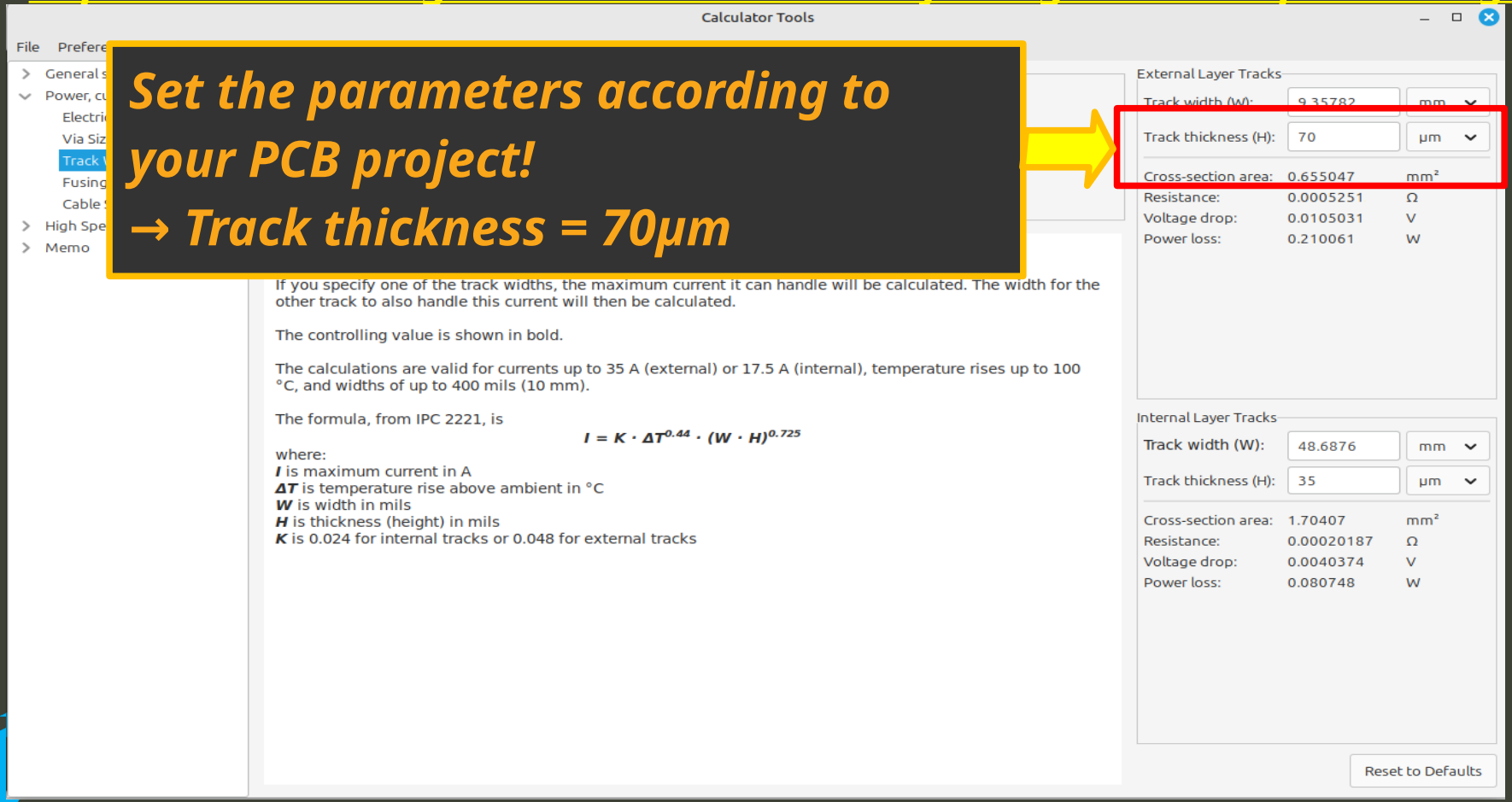
- ***Repetition: → Using KiCad calculator tool for high-current path design***



Set the parameters according to your PCB project!
→ fuel cell max. current (20A)
→ max. track temperature rise (10°C)
→ conductor length

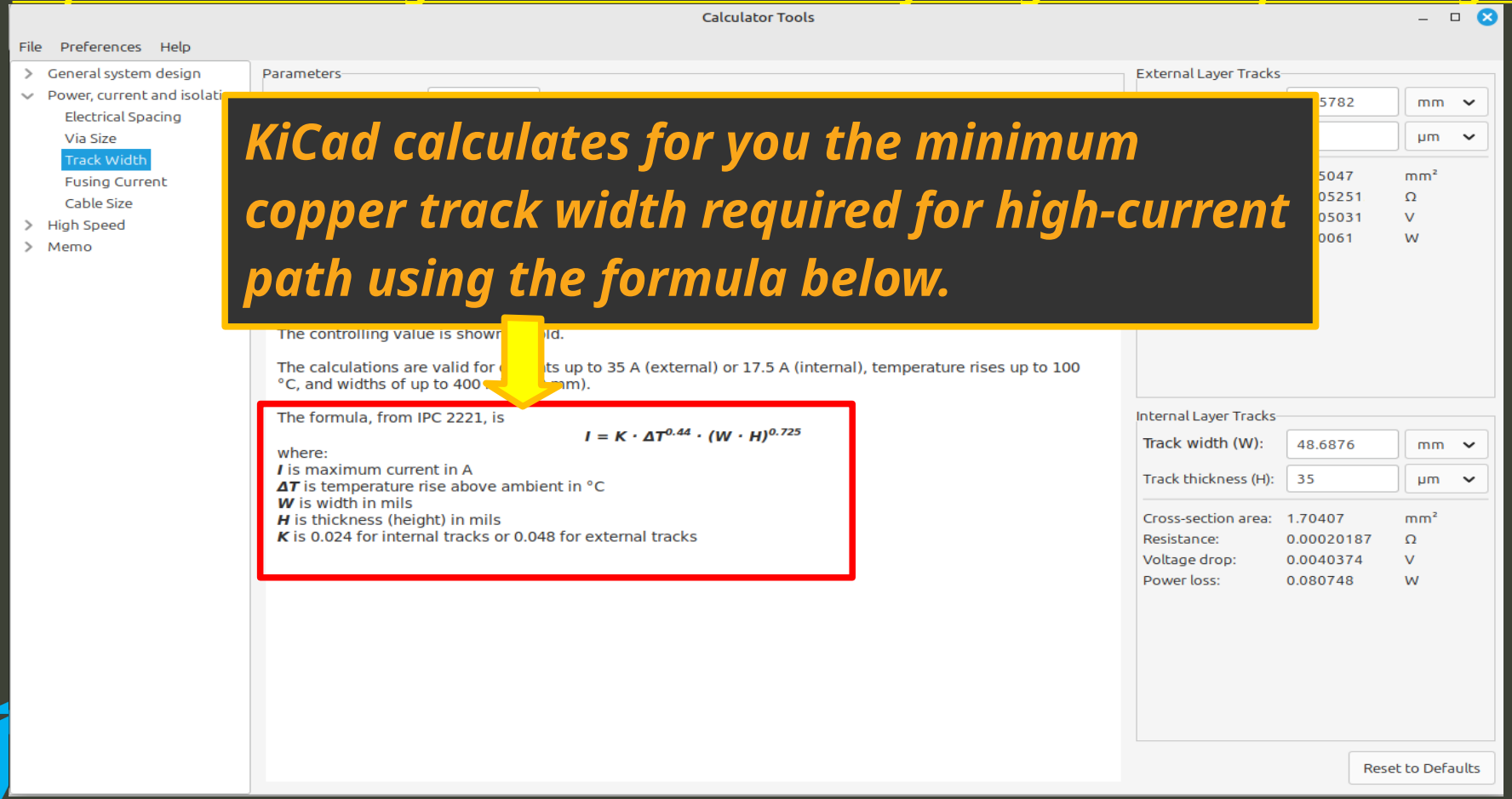
Project Seminar – Project Work

- ***Repetition: → Using KiCad calculator tool for high-current path design***



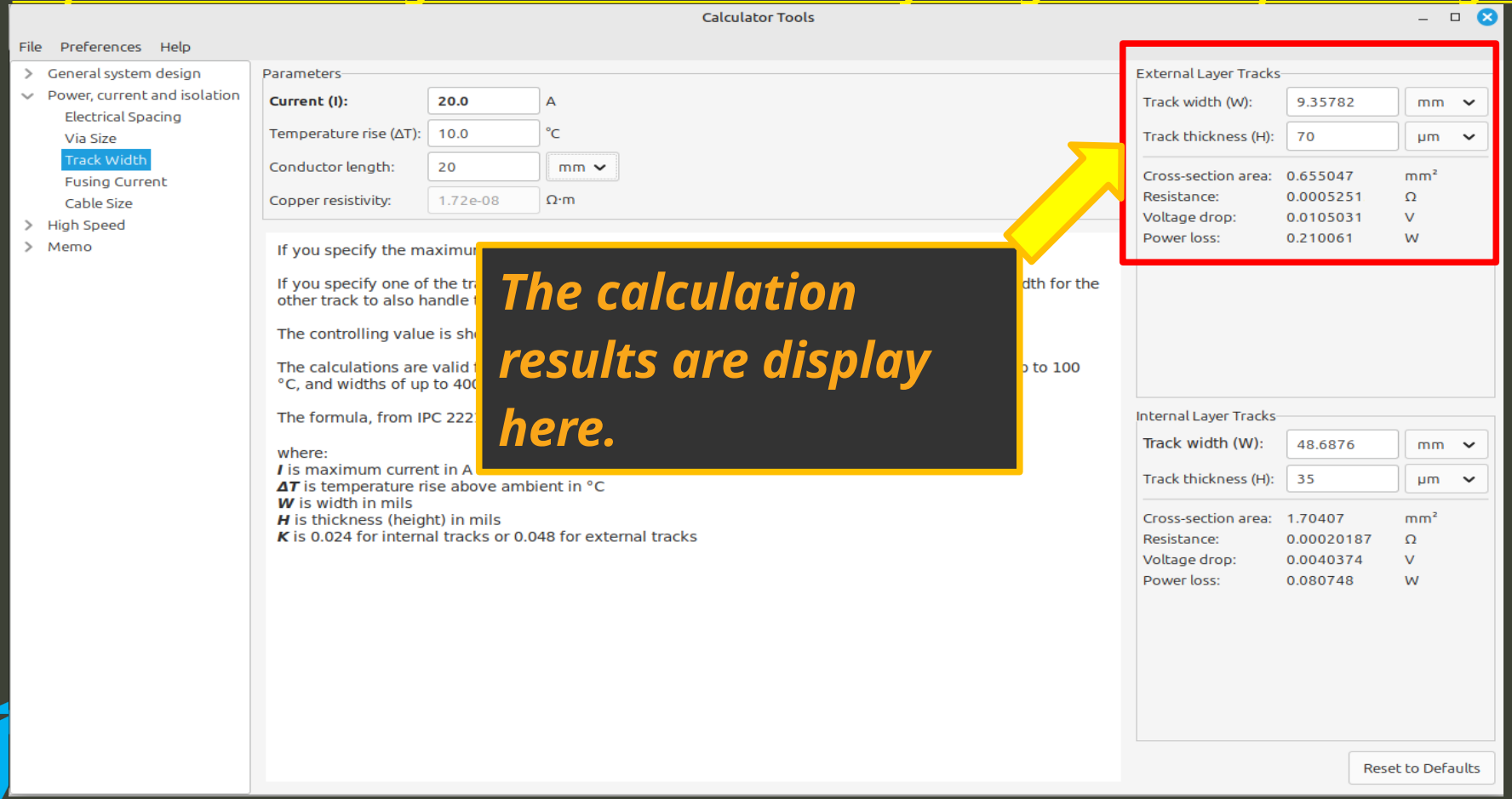
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- ***Repetition: → Using KiCad calculator tool for high-current path design***



Project Seminar – Project Work

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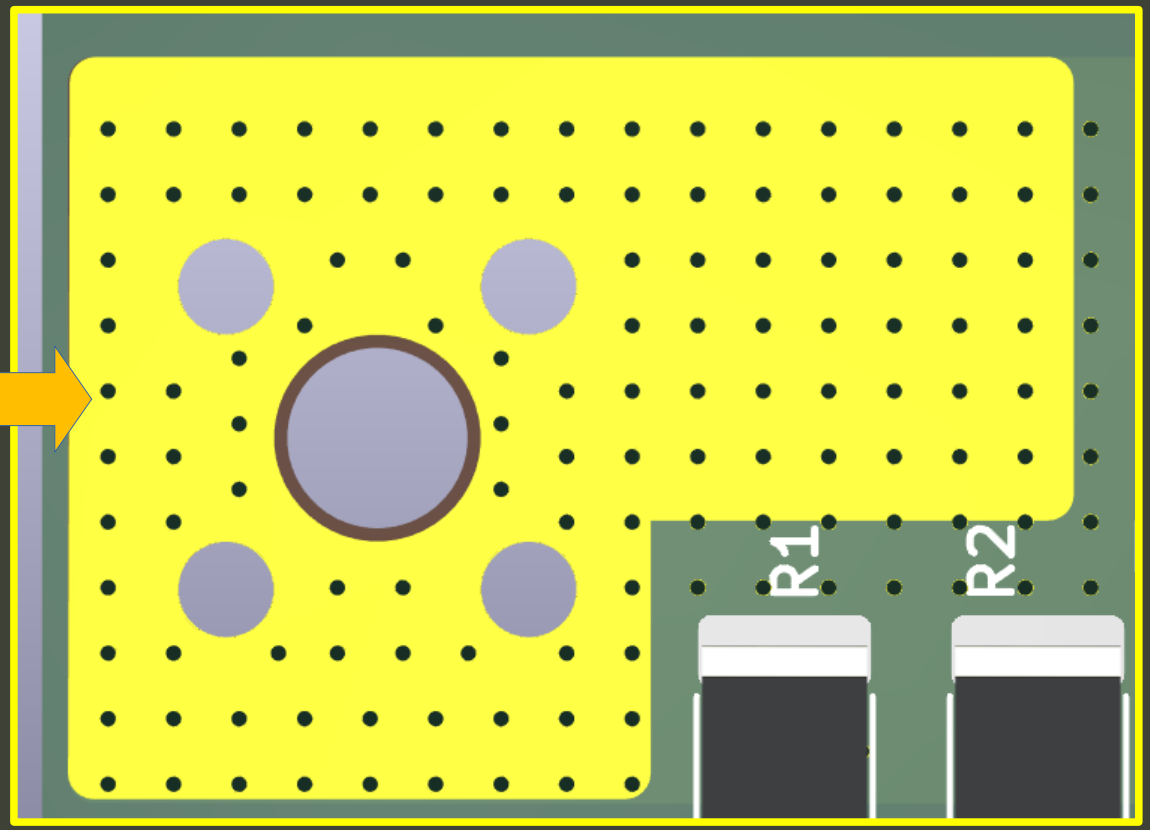


Project Seminar – Project Work

- ***Repetition: High-current path design → result in PCB 3D-viewer***

→ High-current path with track thickness of 70µm and track width of 14mm

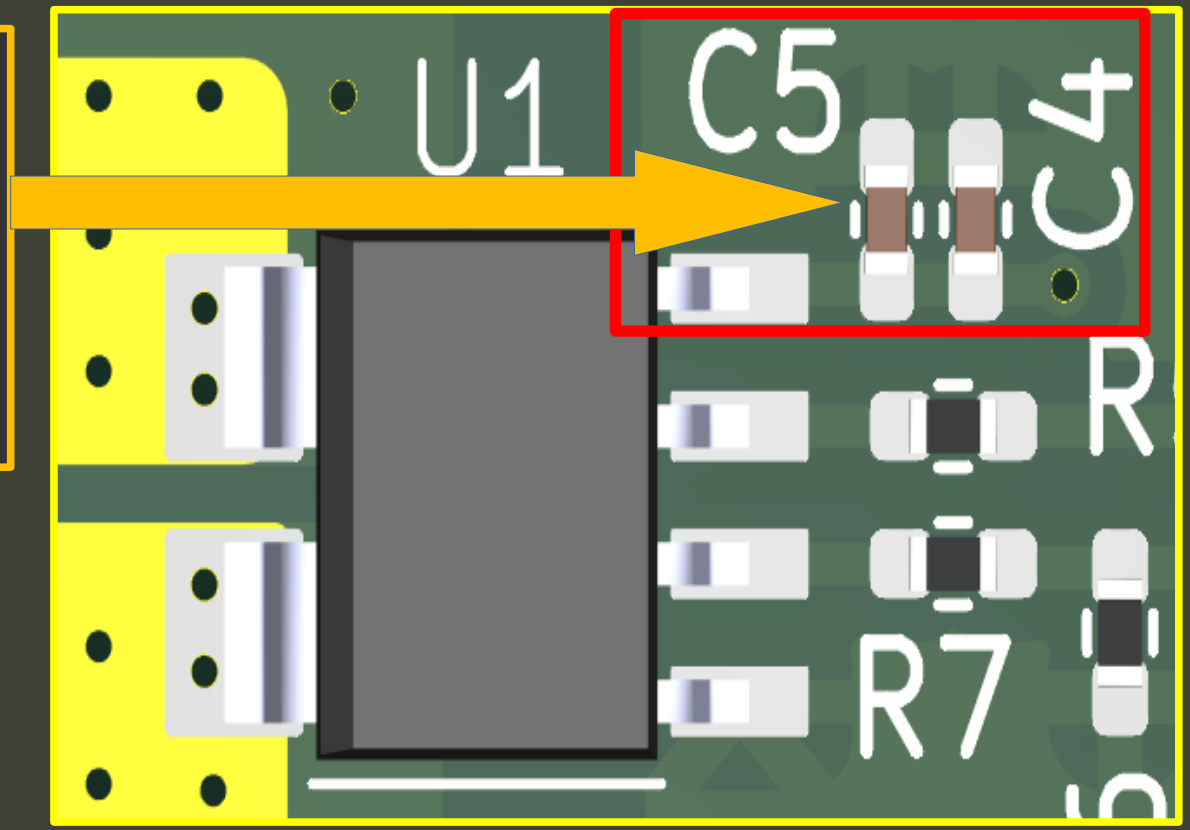
→ solder mask has been removed on the top layer and on the bottom layer



Project Seminar – Project Work

- ***Repetition: Placement of capacitors at supply pins***

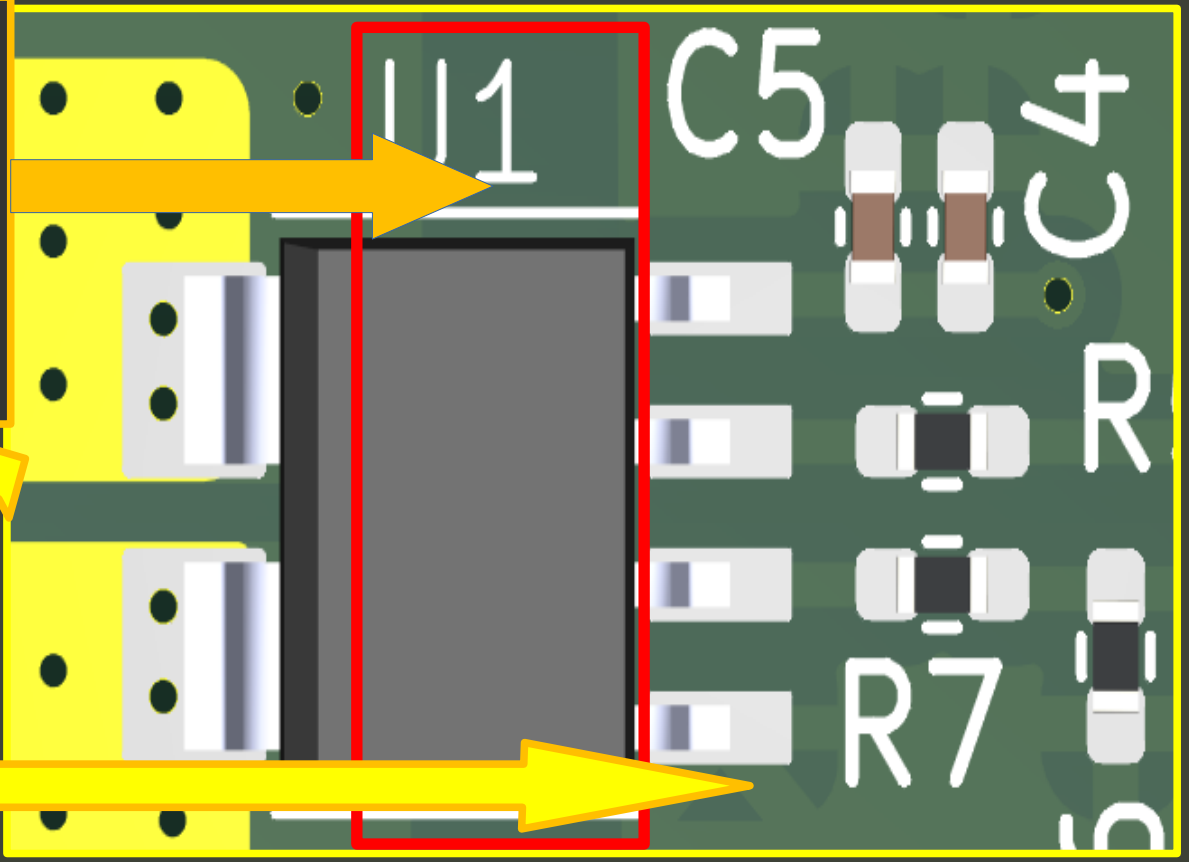
→ Place the bypass capacitors and buffer capacitors as close as possible to the supply pin!



Project Seminar – Project Work

- ***Repetition: Galvanic isolation → hall-effect current sensor***

→ Keep the isolation between the high-current path and analog signals and GNDA on all layers!



High-current path

Analog GND (GNDA)

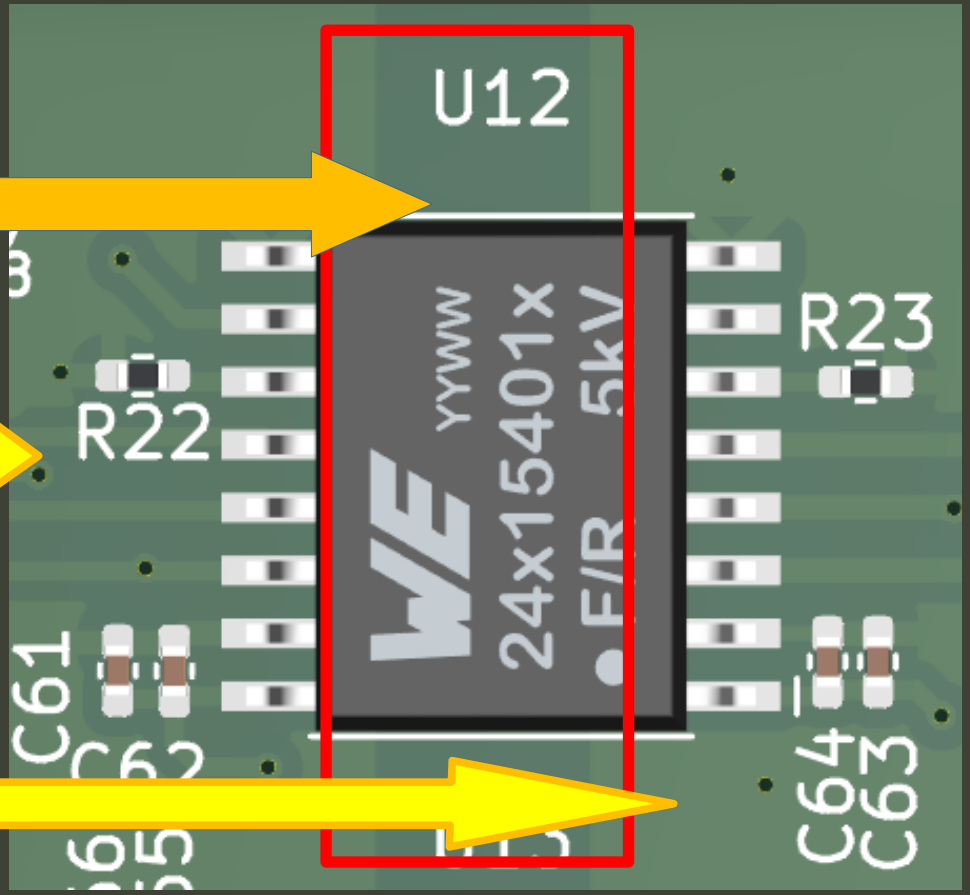
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- ***Repetition: Galvanic isolation → Digital isolators***

→ Keep the isolation between the analog signals and digital signals on all layers!

Analog side

Digital side



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Project Seminar - Schedule



- ***Lessons and tasks:***
  - 28.05.2026: independent study – study complete PCB design
  - until 04.06.2026:
  - → Double-check your PCB in accordance with the PCB design rules
    → Run the DRC in the PCB editor! (review and resolve all errors and warnings!)
    → Submit the current version of your complete KiCad project (including symbols, footprints, schematics and PCB) as a ZIP file.
  - until 04.06.2026: Exercise 8 – How to create a symbol library in KiCad?
    → Describe concisely but in detail how to create a symbol library for a component using the corresponding datasheet and the symbol editor in KiCad!
  - until 04.06.2026: Exercise 9 – How to create a footprint library in KiCad?
    → Describe concisely but in detail how to create a footprint library for a component using the corresponding datasheet and the footprint editor in KiCad!

Project Seminar - Schedule



- ***next lessons:***
  - 05.06.2026: Lesson 9 – Development of operational board software

… schedule is continously updated!

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Thank you !
Have a nice weekend!



